# Bayesian Inference for A/B Testing:

# An Interpretable Alternative to Frequentist Hypothesis Testing

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## Abstract

A/B testing is a fundamental technique used in data-driven decision making to evaluate the effectiveness of competing alternatives. Currently, such experiments are analyzed using frequentist statistical methods, most commonly null hypothesis significance testing and p-values. While widely embraced, these approaches often produce results that are difficult to interpret for decision-making in real life and may reduce experimental outcomes to binary significance judgements. As experimentation becomes increasingly central to product development, marketing optimization, and online platforms, there is growing interest in statistical frameworks that provide more intuitive and precise interpretation of experimental data.

This paper investigates the use of Bayesian inference as an alternative framework for analyzing A/B testing experiments. Using a controlled experimental simulation of only two variants with observed user conversion data, posterior distributions of conversion rates are estimated under a Beta-Binomial model with non-informative priors. Monte Carlo sampling in also employed to calculate the posterior probability that one variant outperforms the other. The results demonstrate that Bayesian methods produce interpretable probabilistic statements about treatment effects while addressing uncertainty through posterior distributions.

In the experimental demonstration, variant B showed a higher observed conversion rate than variant A. Bayesian analysis estimated a posterior probability of approximately 96% that variant B has a higher true conversion rate. Unlike frequentist hypothesis testing, which provides evidence against a null hypothesis without quantifying the probability of competing alternatives, the Bayesian framework directly answers the decision-relevant question of which variant is more likely to perform better.

**Bayesian Inference for A/B Testing: An Interpretable Alternative to Frequentist Hypothesis Testing**

A/B testing is a primary experimental methodology used to compare two alternative models or system designs by measuring user behavior and its impact. It is commonly applied across domains such as web optimization, digital marketing, product development, where small changes in functionality can have important effects on key performance metrics.

In its most basic form, A/B testing consists in randomly assigning users to two variants, commonly known as A (control) and B (treatment), and noticing differences in outcomes such as click-through rates, conversion rates, or engagement metrics.
The most used rate in this domain is the conversion rate or CVR. This rate is the percentage of users who complete a certain action, measuring how campaigns turn audiences into active customers.

The practical importance of A/B testing lies in its power as a decision-making tool while facing uncertainty.
Organizations rely on experimental results to determine whether a change or initiative should be deployed, iterated upon, or completely discarded. These decisions often come with economic consequences, making it of great value not only to detect differences between variants but also to quantify uncertainty and risk in a confident way.
As a result, the statistical framework underlying A/B testing strictly influences the quality and interpretability of the outcomes drawn from experimental data.

Historically, A/B testing, has been dominated by frequentist hypothesis testing methods, considering t-tests, z-tests, and null hypothesis significance testing. These approaches normally

focus on evaluating whether observed data are sufficiently unlikely under a null hypothesis of no effect, using p-values and predefined significance thresholds, which might mean losing flexibility and precision. While such methods provide a mathematically rigorous mechanism for controlling error rates, they present notable flaws in applied environments.

In particular, p-values do not explicitly express the probability that one variant outperforms another, nor do they quantify the size of the effect in probabilistic terms.

Then, we are left out with a disconnection between statistical output and practical interpretation (applicable results) that often leads to confusion, misuse, or overreliance on arbitrary significance thresholds.

This is when Bayesian inference comes in handy. It offers an alternative framework that directly addresses these challenges by modeling uncertainty directly and producing probabilistic statements about quantities of interest. Rather than testing a null hypothesis, Bayesian A/B testing calculates posterior distributions over conversion rates and enables direct comparisons between variants. This allows practitioners to answer intuitive and more natural questions such as the probability that variant B outperforms variant A, or the likelihood that an observed change exceeds a significant threshold.

Such outputs are instinctively aligned with real-world decision-making processes.

This paper argues that Bayesian inference can provide a more interpretable and decision-focused framework for A/B testing than traditional frequentist methods. Through a practical implementation using Beta-Binomial modeling and Monte Carlo sampling, we demonstrate how Bayesian methods contribute to clearer insights into uncertainty, effect size, and mostly risk.

A comparative analysis highlights the way Bayesian thinking better encourages experimental

evaluation in applied data science settings, specifically when decisions must be made under the unknown and incomplete information.

**Background and Related Work:**

***Frequentist Approaches to A/B Testing***

Frequentist statistical methods have long served as the predominant framework for A/B testing and controlled experimentation. In this paradigm, outcomes are categorized as realizations of random variables under the influence of fixed but unknown parameters, and inference is performed through hypothesis testing procedures.
Commonly used methods or techniques include two-sample t-tests, z-tests for proportions, and chi-squared tests, depending on the base of the outcome variable and sample size assumptions (Casella & Berger, 2002).

In a typical frequentist A/B test, a null hypothesis is used $H_0$, stating that there is absolutely no difference between variants A and B. Statistical significance is then assessed by computing a test statistic and its p-value, which serves as a representation of the probability of observing data at least as extreme as the observed sample, assuming the null hypothesis is true. If the p-value exists below a predetermined significance threshold (commonly $\alpha = 0.05$), the null hypothesis is rejected in favor of an alternative hypothesis $H_a$.

As it was mentioned before, this framework indeed provides an exact rigorous mechanism for controlling Type I error rates, but unfortunately its interpretation is often misunderstood or misapplied in practice.

As emphasized by Wasserstein and Lazar (2016), p-values do not measure the probability that a hypothesis is true, nor do they quantify the magnitude or practical relevance of an effect. Instead, they reflect the mere compatibility of the observed data with a specific null model.

In applied A/B testing contexts, these impediments can be problematic. Decisions are frequently framed in terms of actionable questions such as whether a new variant is better than the current one, or even how confident one can be in deploying a change.
Frequentists outputs, however, typically reduce such decisions to a binary outcome, reject or fail to reject the null hypothesis, without explicitly expressing uncertainty in terms that are applicable to the real world.
Moreover, reliance on fixed significance thresholds introduces arbitrariness and can obscure meaningful but unprecise and uncertain effects, particularly in experiments with limited sample sizes or small effect sizes.

**Bayesian Inference and Probabilistic Modeling**

Bayesian statistics offer an alternative inferential framework based in probability theory as a way of measuring a belief rather than a long-run frequency.
In the Bayesian paradigm, uncertainty about unknown parameters and data is represented explicitly through probability distributions, which are updated as new data are observed according to Bayes' theorem (Gelman et al., 2013).

Now, formally, Bayesian inference mixes three key components: a prior distribution that takes initial beliefs about the parameters of interest, a likelihood function capable of modeling the data-generating process, and finally a posterior distribution that represents updated beliefs

after being fed with data. This posterior distribution sets up the basis for all subsequent inference, including point estimates, uncertainty intervals, and probabilistic comparisons.

One of the main advantages of Bayesian inference is its ability to be interpreted. Posterior probabilities can be directly interpreted as degrees of belief about parameter values or hypotheses, allowing statements such as “there is a 95% probability that the conversion rate of variant B exceeds that of variant A”.
Such declarations are not possible within the frequentist framework without additional assumptions or approximations.

Bayesian methodologies have seen increasing adoption in applied fields such as machine learning, decision theory, and experimental design, especially in settings where uncertainty quantification and sequential learning are critical (Murphy, 2012).

In contrast to fixed-sample hypothesis testing, Bayesian models by nature support incremental updating as new data appears, making them well-suited for modern experimentation workflows.

**Bayesian A/B Testing and the Beta-Binomial Model**

In the context of A/B testing with binary outcomes (e.g., conversion versus non-conversion), the Beta-Binomial model is a canonical Bayesian formulation. User outcomes are modeled as Bernoulli trials governed by an unknown conversion chance, while uncertainty over this probability, while uncertainty over this probability is represented by a Beta distribution. The Beta distribution is particularly appealing due to its role as a conjugate prior for the Binomial likelihood, yielding analytically tractable posterior updates (Bernardo & Smith, 2000).

Under the lens of this model, studying conversion data results in a posterior distribution whose parameters are constantly updated by simple additive rules, enabling efficient inference even in large-scale experimentations. The resulting posterior distribution traps both the estimated conversion rate and the related uncertainty, which can be visualized and summarized using credible intervals or future density plots.

Bayesian A/B testing specifically based on the Beta-Binomial framework has been advocated throughout the years as a more intuitive alternative to traditional old-school hypothesis testing, particularly in industry application. As noted by McElreath (2020), Bayesian models allow practitioners to reason directly about quantities they are interested in, rather than indirect test statistics.

This perspective has contributed to picking interest in Bayesian experimentation platforms within technology companies and data-driven organizations.

**Monte Carlo Methods for Posterior Inference**

While we can admit that some Bayesian models admit closed-form posterior resolutions, many practical inference tasks require numerical approximation.

Monte Carlo methods do provide a general class of techniques for approximating in expectations and probabilities by random sampling.

In Bayesian inference, Monte Carlo sampling is key to approximate posterior distributions and derived quantities when analytical responses are impractical or unavailable (Robert & Casella, 2004).

In simple conjugate models such as the Beta-Binomial case, direct sampling from the posterior distribution is feasible and most importantly computationally efficient.

Drawing random samples from the after-results allows practitioners to estimate probabilities of interest (such as the probability that one variant confidently performs better than another) by empirical frequencies over sampled values.
This sampling-based context transforms abstract probabilistic inference into a concrete simulation process that is both intuitive and very flexible.

Monte Carlo techniques play a foundational role in modern Bayesian computation, breaking the barriers of simple sampling based on randomness to more advanced methodologies such as Markov Chain Monte Carlo (MCMC) and variational inference.
Although such tools are not required for the present analysis, their widespread use underscores the scalability and malleability of Bayesian inference across increasingly complex models.

## Method

### Experimental Setup

We take in consideration a standard A/B testing scenario in which users are randomly assigned to one of two experimental variants: a control version (A) and a treatment version (B). Each user interaction produces a binary outcome indicating whether a conversion happened. Let $Y_i \in \{0,1\}$ denote the result for the $i$-th user, where 1 stands for a conversion and 0 represents a non-conversion.

The main objective of the experiment is to compare the underlying conversion probabilities $\theta_A$ and $\theta_B$ associated with variants A and B, respectively, while direclty accounting for uncertainty in their estimation.

To guarantee reproducibility and interpretability, both simulated data and analytical inference procedures are applied. Simulation aims for controlled experimentation with known parameters, while inference methods are evaluated based on their ability to support decision-making under uncertainty.

**Data Generation Process**

User conversion results are modeled as independent Bernoulli trials. For each variant $v \in \{0,1\}$, outcomes are created according to: $Y_i^{(v)} \sim \text{Bernoulli}(\theta_v)$ where $\theta_v$ represents the true (but unknown) conversion probability for variant $v$.

In the experimental demonstration, a fixed number of users is assigned to each variant, and outcomes are generated with predefined conversion probabilities.
The simulation-based proposal ensures a known ground truth against which inferential methods can be fairly compared.

**Pseudocode: Data Generation**

This procedure implements binomial sampling, where each Bernoulli trial represents a single user interaction.

**Frequentist Inference Procedure**

As a starting comparison, a frequestist hypotheis testing method is aplied to the observed data. Let $\bar{Y}_A$ and $\bar{Y}_B$ denote the empirical conversion rates for variants A and B. A two-sample test is performed under the following null hypothesis:

$H_0: \theta_A = \theta_B$

against the alternative hypothesis:

$H_1: \theta_A \neq \theta_B$

*A* test statistic is computed based on the difference in sample means, and an equivalent p-value is obtained. Statistical significance is assessed using an established significance level $\alpha$, commonly set to 0.05.

The outcome of this procedure is a binary decision to reject or fail to reject the null hypothesis. While this methodology provides a standardized testing framewrok to work with, it does not yield an explicit probabilistic statement regarding the dominance of one variant over the other.

**Bayesian Model Specification**

***Likelihood***

In the Bayesian framework, the same Bernoulli likelihood is implicit for user outcomes:

$$P(Y \mid \theta_v) = \prod_{i=1}^{n_v} \theta_v^{Y_i} (1 - \theta_v)^{1-Y_i}$$

Where $n_v$ is the number of users assigned to variant $v$

**Prior Distribution**

Uncertainty over the conversion probabilities $\theta_A$ and $\theta_B$ is modeled using Beta prior distributions:

$\theta_v \sim Beta(\alpha_0, \beta_0)$

A non-informative prior is chosen by setting $\alpha_0 = \beta_0 = 1$, corresponding to an uniform distribution over the interval [0,1]. This selection reflects an absence of prior preference for any particular conversion rate while safeguarding mathematical stability.

**Posterior Distribution**

Even though the posterior distributions are analytically tractable, direct analytical comparison between variants is nontrivial. Instead, Monte Carlo Sampling is implemented to approximate posterior quantities of interest.

For eacgh variant, a large number of samples is drawn from the corresponding posterior distribution:

$$\theta_v^{(1)}, \theta_v^{(2)}, \dots, \theta_v^{(M)} \sim Beta(\alpha_v, \beta_v)$$

Where *M* denotes the number of samples.

These samples are plausible realizations of the true conversion rate, weighted according to posterior belief.

**Evaluation Metrics**

The primary Bayesian evaluation metric used in this study is the probability of superiority:

$$P(\theta_B > \theta_A)$$

Which is estimated empirically from posterior samples. Unlike frequentist significance tests, this metric direclty answers the question of interest: how likely it is that variant B outperforms variant A.

Secondary summaries consider posterior means, credible intervals, and visual inspection of posterior density functions.

**Visualization of Posterior Distributions**

Posterior distributions are visualized by evaluating the probability density function over a grid of plausible coversion rates and plotting the resulting curves. This plot (Figure 1) provides insight into both effect size and uncertainty.

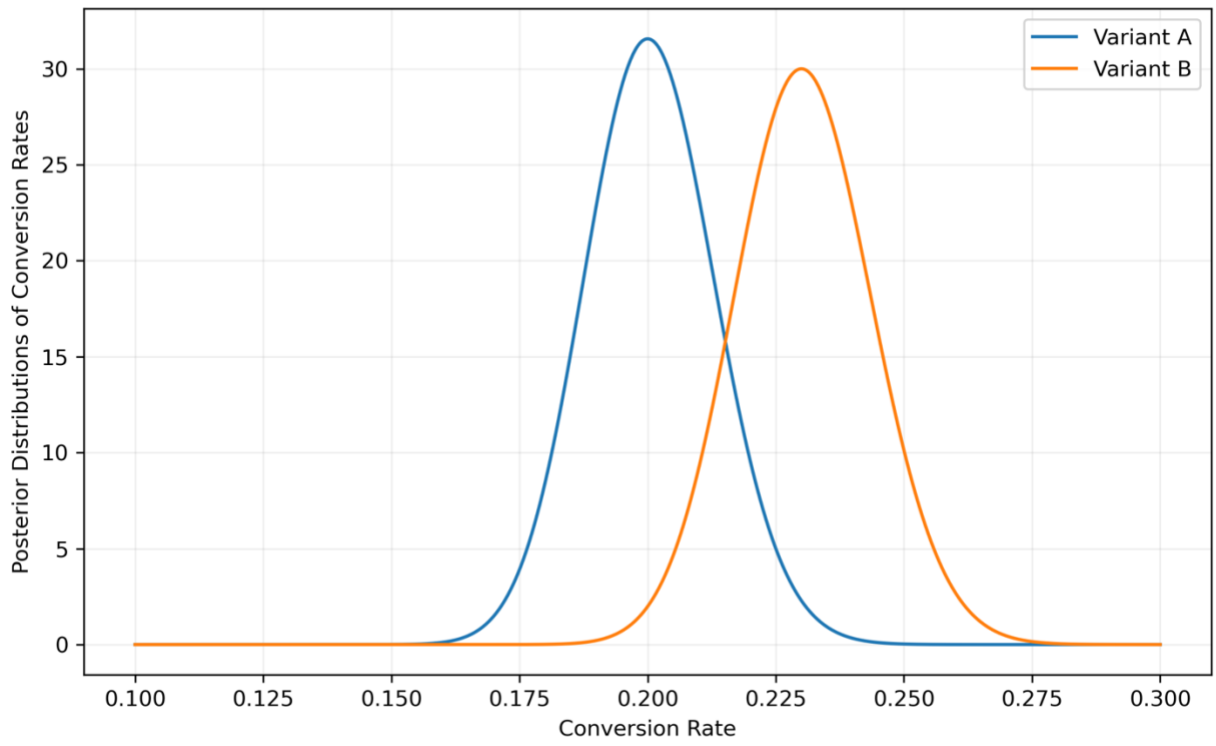


*Figure 1*

### Reproducibility and Implementation

All experiments were implemented in Python using standard scientific computing libraries, including NumPy, Pandas, SciPy, and Matplotlib. Random seeds were fixed to ensure reproducibility. The full codebase, including data generation, inference procedures, and visualization scripts, is provided un an accompanying appendix.

## Results

This section presents the outcomes of the Bayesian A/B testing framework applied to the experimental data and opposes the resulting insights with those obtained through frequentist hypothesis testing. The focus is placed on interpretability, uncertainty quantification, and decision relevance.

### Posterior Distributions of Conversion Rates

Figure 1 shows the posterior distributions of the conversion rates for variants A and B after observing 1000 user interactions per variant. Variant A exhibits 200 conversions, corresponding to an empirical conversion rate of 0.20, while variant B exhibits 230 conversions, corresponding to an empirical conversion rate of 0.23.
Under a Beta(1,1) prior, these observations yield posterior distributions Beta(201, 801) for variant A and Beta(231, 771) for variant B.

The posterior distributions show both the estimated conversion rates and the uncertainty associated with each estimate. Variant B's posterior distribution is moved to the right relative to variant A, indicating a higher expected conversion rate.
However, a non-negligible overlap between the distributions remains, highlighting residual

uncertainty despite the observed difference. This overlap visually communicates the level of confidence in the treatment effect without reducing the result to a pure binary outcome.

**Monte Carlo Estimation of Probability of Superiority**

To quantify the likelihood that variant B is in fact better than variant A, Monte Carlo sampling was performed by drawing 10000 samples from each posterior distribution.
For each sampled pair of conversion rates, the condition $\theta_B > \theta_A$ was evaluated. The posterior probability of superiority was then calculated as the proportion of samples satisfying this condition.

The resulting estimate states that variant B outperforms variant A with a posterior probability of approximately 0.96. This value can be read direclty as the probability, given the observed data and model assumptions, that the true conversion rate of variant B exceeds that of variant A.
Unlike frequentist significance testing, this result has continuity and an interpretable measure of confidence rather than binary decision based on an arbitrary threshold.

**Comparison with Frequentist Results**

Under a frequentist framwork, the same data yield a statistically significant difference between variants when evaluated using a two-sample test for proportions.
While this result does support rejecting the null hypothesis of no difference, it does not directly quantify the probability that variant B is better than variant A, nor does it express uncertainty in meaningful terms for decision-making.

In contrast, the Bayesian results explicitly address uncertainty through posterior distributions and provide a probability of superiority that answers how likely it is that one variant truly outperforms the other.

## Discussion

The result of this study highlights main differences between Bayesian and frequentist approaches to A/B testing, specifically with respect to interpretability, uncertainty representation, and decision support. While both frameworks identify a performance difference between variants A and B, the nature of the conclusions they enable differs substantially.

A key advantage of the Bayesian framework is its ability to represent uncertainty explicitly through posterior distributions. Rather than summarizing results through a single test statistic or p-value, Bayesian inference produces a full distribution over plausible values of the conversion rate. This allows practitioners to reason not only about the expected performance but also about the range and likelihood of alternative outcomes.
In practical experimentation environments, where uncertainty is inherent and decisions must be made with incomplete data, such representations provide valuable context that is absent from traditional hypothesis testing.

The posterior probability of superiority further illustrates the decision-oriented nature of Bayesian inference. Estimating the probability that variant B outperforms variant A implies a direct and interpretable quantity that aligns closely with the questions faced by practitioners. Unlike frequentist significance testing, which evaluates the compatibility of data with a null hypothesis, Bayesian inference addresses the probability of competing hypotheses given the analyzed data. This distinction is particularly important in applied settings, where decisions are

rarely framed in terms of null hypothesis rejection but instead are based on assessing a risk, confidence, and expected benefit.

Also, another important thing to consider is the reliance of frequentist methods on fixed significance thresholds. While such thresholds are useful for controlling error rates in repeated sampling, they carry an element of arbitrariness than can obscure meaningful effects, especially in experiments with important sample sizes. Bayesian inference avoids this problem by treating uncertainty as a continuous quantity rather than a binary outcome. Then, evidence can be accumulated and interpreted gradually, supporting more nuanced conclusions.

Despite these advantages, Bayesian methods are not free from limitations. Posterior inferences depend on modeling assumptions, including the choice of prior distribution and the specification of the likelihood.

Although a non-informative prior was used in this study to minimize subjective influence, different prior choices could lead to variations in posterior approximations, particularly in low-data regimes. Additionally, the current analysis assumes independent user interactions and a simple binary outcome, which may not fully contain the complexity of real-world user behavior.

Future work could extend this framework by incorporating informative or hierarchical priors, modeling temporal effects, or transforming the approach to multi-variant experiments and continuous outcomes.

Sequential decision-making strategies, such as Bayesian adaptive experimentation, also state promising directions for further investigation. These extensions would posteriorly enhance the applicability of Bayesian A/B testing to large-scale, dynamic experimentation environments.

Overall, the findings of this study suggest that Bayesian inference offers a more transparent and reliable decision-relevant framework for A/B testing than traditional frequentist

methods. By enabling direct probabilistic reasoning and explicit uncertainty quantification, Bayesian approaches align more closely with the practical needs of data-driven decision-making in modern applied settings.

## Appendix

```
import numpy as np

# True (unknown) conversion probabilities
theta_A = 0.12  # version A converts 12% of the time
theta_B = 0.15  # version B converts 15% of the time

# Number of users per variant
n_users = 100

# Generating conversion outcomes
Y_A = np.random.binomial(1, theta_A, n_users)
Y_B = np.random.binomial(1, theta_B, n_users)

# Observed conversion rates
conversion_rate_A = Y_A.mean()
conversion_rate_B = Y_B.mean()

conversion_rate_A, conversion_rate_B
```

**Pseudocode: Posterior Sampling and Comparison**

```
import numpy as np

def posterior_superiority(alpha_A, beta_A, alpha_B, beta_B,
M=100_000):

# Draw samples from posterior distributions
samples_A = np.random.beta(alpha_A, beta_A, M)
samples_B = np.random.beta(alpha_B, beta_B, M)

# Estimate probability that B is better than A
prob_B_better_than_A = np.mean(samples_B > samples_A)

return samples_A, samples_B, prob_B_better_than_A
```

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import beta

np.random.seed(42)

# Observed Data (Canonical Experiment)
n_a = 1000
n_b = 1000

a_success = 200
a_fail = 800

b_success = 230
b_fail = 770

# Posterior Distributions
posterior_a = beta(1 + a_success, 1 + a_fail)
posterior_b = beta(1 + b_success, 1 + b_fail)

# Grid
x = np.linspace(0.1, 0.3, 1000)

# Plot posterior PDFs
plt.figure(figsize=(8, 5))
plt.plot(x, posterior_a.pdf(x), label='Variant A')
plt.plot(x, posterior_b.pdf(x), label='Variant B')
plt.xlabel('Conversion Rate')
plt.ylabel('Posterior Distributions of Conversion Rates')
plt.legend()
plt.tight_layout()
plt.grid(alpha=0.2)
plt.savefig('posterior_distributions.png', dpi=300)
plt.show()

# Monte Carlo Sampling
samples_a = posterior_a.rvs(10000)
samples_b = posterior_b.rvs(10000)

# Probability of superiority
prob_b_better = np.mean(samples_b > samples_a)
prob_b_better
```

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